

Deploy LPWAN sensors for energy optimization in traditional manufacturing plants undergoing digital transformation.

Manufacturing × Energy optimisation × LPWAN (LoRaWAN, NB-IoT, LTE-M) · OS161 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
69 is the world moving	69 can we play, can we win	MEDIUM 0 named in the evidence	€2.9bn partial evidence · per year	NEXT pilots published no volume pr...	L0 17 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity is to deploy LPWAN sensors for energy optimization in traditional manufacturing plants that are undergoing digital transformation. It targets manufacturers in sectors like equipment manufacturing and food processing, where energy inefficiency and sustainability pressures are driving the adoption of sensor-based monitoring and optimization. The time is right because these plants are already shifting toward intelligent operations and green transformation, as evidenced by recent industry reports.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€4.0bn €504m – €26.0bn	€2.9bn €362m – €18.6bn	€99.8m €12.7m – €653m	partial
Observed: tendered contract value, annualised	€20.9m €20.9m – €56.6m	€20.9m €20.9m – €56.6m	€733k €733k – €2.0m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Manufacturing (EU27_2020)	238,122 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting LPWAN (LoRaWAN, NB-IoT, LTE-M)	30.0 % of enterprises (10+ employees)	proxy	Eurostat · isoc_eb_iotn2 · E_IOT1	2021
Annual value of one engagement	€337k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Size-weighted buyer base	39,220 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	3.5 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Internet of Things by NACE Rev. 2 activity series E_IOT1 is used as a proxy for LPWAN (LoRaWAN, NB-IoT, LTE-M): IoT adoption as the connectivity demand proxy. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders. 2 of 3 declared geographies are outside the European reference data (AU, CN); the estimate covers the European footprint only.

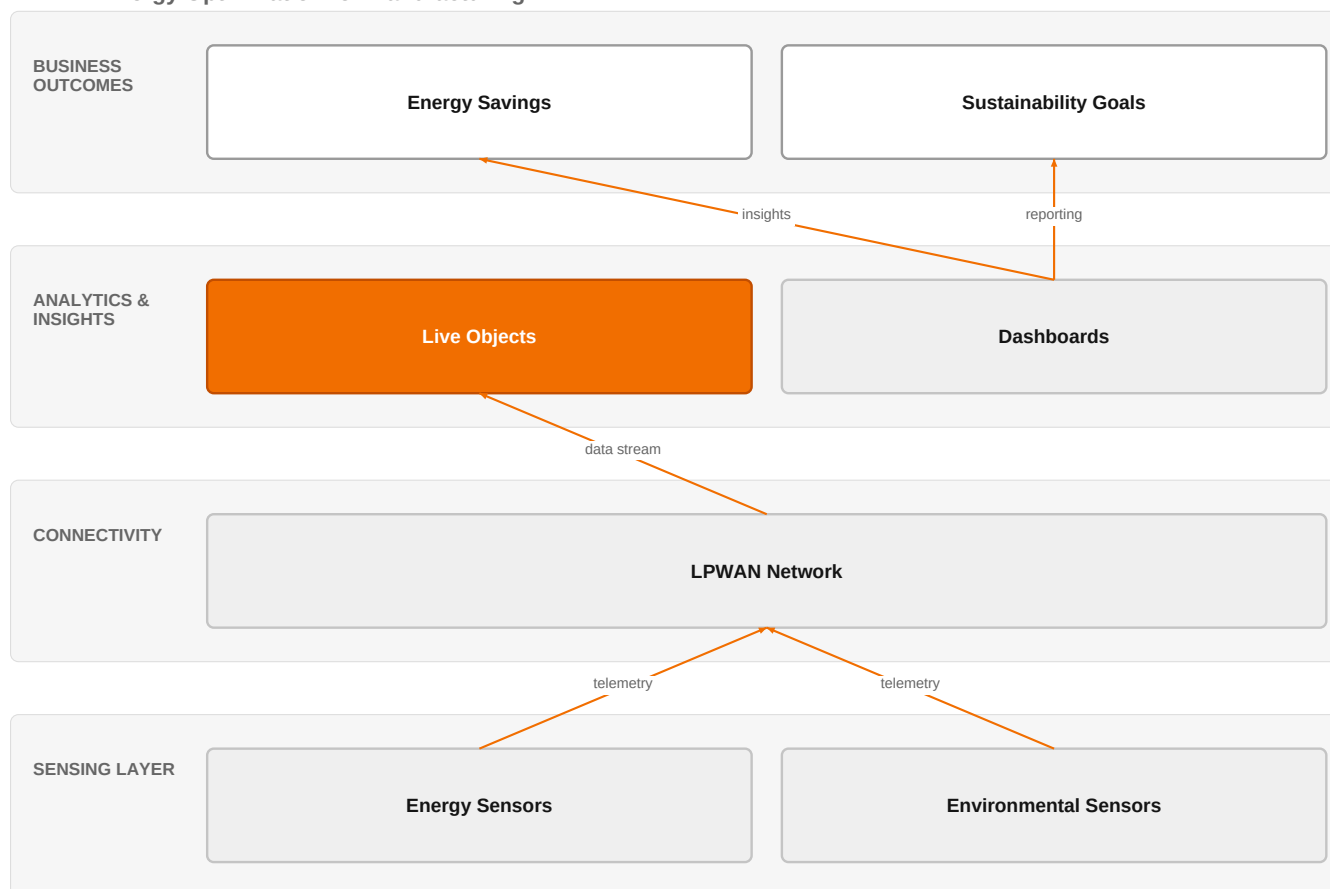
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€20.9m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-05
Assumed obtainable share of the serviceable market	3.5 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 14 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size. 2 of 3 declared geographies are outside the European reference data (AU, CN); the estimate covers the European footprint only.

The solution, in one picture

LPWAN Energy Optimization for Manufacturing



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This diagram shows how LPWAN sensors feed data through Live Objects to deliver actionable insights for energy savings and sustainability.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

Traditional manufacturing plants are increasingly adopting digital technologies, as seen in the transformation of turbine factories into intelligent facilities. Regional initiatives, such as Shanxi's promotion of green transformation in equipment manufacturing, are pushing energy optimization to the forefront. In mining, reliable connectivity is identified as a foundational challenge for decarbonisation, implying a need for sensor networks. Additionally, frameworks like Green HACCP are extending traditional practices to address energy inefficiency, signaling a broader trend toward sustainability in manufacturing.

Sources: SIG-14947CE171FF stdaily.com 2026-08-14; SIG-1B97CEF269AB en.people.cn 2026-08-11; SIG-5CEFDDFAA80A mining-technology.com 2026-08-14; SIG-7697DB6F38B5 openalex.org 2025-08-30

Who buys, and why now

The buyer is typically the plant manager or operations director, who feels the pain of rising energy costs and sustainability targets. The trigger is often an energy audit or a regulatory push that forces the plant to reduce consumption. They need to see real-time data on energy usage across the facility to identify waste and optimize processes, but they lack the visibility that LPWAN sensors can provide.

Why it is hot now — every claim bound to a dated source

- Traditional turbine factories are turning intelligent, indicating a shift towards digital technologies in manufacturing.

[SIG-14947CE171FF]

- Shanxi province promotes green transformation of equipment manufacturing, which includes energy optimization initiatives. [SIG-1B97CEF269AB]
- The mining decarbonisation challenge starts with reliable connectivity, implying sensor networks. [SIG-5CEFDFAA80A]
- Mining shows showcase technologies shaping the industry's future. [SIG-1E1C81947E78]
- Green HACCP extends traditional HACCP to address environmental sustainability, including energy efficiency. [SIG-7697DB6F38B5]
- Green HACCP integrates Environmental Respect Practices to address excessive water use and energy inefficiency in food safety management. [SIG-7697DB6F38B5]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution using its IoT experts and Live Objects platform to connect LPWAN sensors across the plant, enabling real-time energy monitoring. The engagement would start with a site assessment, then deploy sensors, connect them via Live Objects, and provide dashboards and analytics to identify optimization opportunities. Orange Group researchers could support with advanced analytics, and the solution would be delivered on ISO 27001 certified infrastructure, with TISAX certification available for automotive plants.

Why Orange can win

Orange can win because it brings a combination of IoT expertise, a proven platform in Live Objects, and certifications like ISO 27001 and TISAX that address trust and security concerns. The references from Colgate-Palmolive, Mondelez International, and Saint-Gobain Glass demonstrate experience in manufacturing environments. However, the assets are not unique, and competitors like Telia and Vodafone also offer LPWAN, so the differentiation must come from the integration and trust aspects.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Live Objects	offer	LO	offer addresses the use case AND provides the technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
TISAX	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Colgate-Palmolive	reference	SUP	published reference in this vertical, different use case	unconfirmed
Mondelez International	reference	SUP	published reference in this vertical, different use case	unconfirmed
Saint-Gobain Glass	reference	SUP	published reference in this vertical, different use case	unconfirmed
Skytale	reference	SUP	published reference in this vertical, different use case	unconfirmed

6 further published reference(s) in this vertical are linked to this topic and are listed in the radar.

Competitive landscape

MEDIUM competitive intensity (score 4.8; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitor	Category	Position here	Basis	Evidence
Telia Company	Telecom operator peer	sells LPWAN (LoRaWAN, NB-IoT, LTE-M); competes in OX: Smart Industries	structural	—
Vodafone Business	Telecom operator peer	sells LPWAN (LoRaWAN, NB-IoT, LTE-M); active in Manufacturing; competes in OX: Smart Industries	structural	—

Activity	Industrial / OT specialist	sells LPWAN (LoRaWAN, NB-IoT, LTE-M); competes in OX: Smart Industries	structural	—
Accenture	Global systems integrator	active in Manufacturing; competes in OX: Smart Industries	structural	—
Capgemini	Global systems integrator	active in Manufacturing; competes in OX: Smart Industries	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	active in Manufacturing; competes in OX: Smart Industries	structural	—
Bosch	Industrial / OT specialist	active in Manufacturing; competes in OX: Smart Industries	structural	—
Dassault Systemes / 3DS Outscale	Sovereign cloud provider	active in Manufacturing; competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 What is your current energy monitoring approach, and where are the biggest gaps in visibility?
- 2 Have you already deployed any IoT sensors or LPWAN networks in your plant?
- 3 What are your specific energy reduction targets, and what is the timeline?
- 4 Who is responsible for energy management, and how are decisions made about capital investments?
- 5 Are you facing any regulatory or customer pressures to reduce your carbon footprint?
- 6 What is your experience with digital transformation initiatives so far, and what has been the biggest challenge?

Objections you will meet

They will say	You can answer
We already have a building management system that tracks energy.	That's true, and a BMS is a good start. However, it often lacks the granularity to identify machine-level inefficiencies. LPWAN sensors can complement your BMS by providing low-cost, easy-to-deploy monitoring across the entire plant, including areas not covered by the BMS.
LPWAN is not reliable enough for our critical operations.	Reliability is a valid concern. That's why we offer multiple LPWAN options, including LoRaWAN, NB-IoT, and LTE-M, and we can design a network with redundancy. Our Live Objects platform also provides robust data management, and we have experience with mission-critical deployments in manufacturing.
The upfront cost of sensors and deployment is too high.	We understand budget constraints. The cost of LPWAN sensors has decreased significantly, and the energy savings can often justify the investment. We can also start with a pilot to demonstrate the value before scaling, and our flexible deployment options can align with your budget cycles.

Risks and unknowns

A key risk is that traditional manufacturers may be slow to adopt new technologies due to legacy systems and resistance to change. The business case depends on energy savings, which can be hard to quantify without a pilot. It is also unknown how quickly the customer's digital transformation will progress, and whether they have the internal skills to maintain the sensor network. The competitive landscape is crowded, and differentiation may be difficult.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for an energy optimization project at a traditional manufacturing plant, highlighting Orange's ISO 27001 and TISAX certifications, and referencing Saint-Gobain Glass, Skytale, and Everise as proof points, while leveraging Live Objects for sensor management.
Sales	In a conversation with a traditional manufacturing plant's operations director, mention how Orange's Live Objects platform, used by Colgate-Palmolive and Mondelez International, can help monitor and reduce energy consumption through LPWAN sensors.
Strategist / Innovator	Commission a deep-dive brief on LPWAN-enabled energy optimization for traditional manufacturing, leveraging Orange's IoT experts and researchers to validate the business case and identify pilot opportunities.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-216688A4B953	2026-03-01	openalex.org	1	Digital Twin-Enabled Thermal Energy Management System for Sustainable Manufacturing Process Opt...
SIG-505D0AD4A3FA	2026-02-13	openalex.org	1	Digital twin model for performance analysis and manufacturing process optimization
SIG-A067578BB3A4	2026-02-06	openalex.org	1	Optimization of the Design and Manufacturing Processes for Metal Additive Manufacturing Through...
SIG-0964410DC575	2025-12-17	openalex.org	1	Generative and Predictive AI for digital twin systems in manufacturing
SIG-6C22EFB6A3A7	2025-12-17	openalex.org	1	On the reusability of machine learning-based process monitoring systems for manufacturing digit...
SIG-436BC624B8F1	2025-11-01	openalex.org	1	A Methodology for Assessing Digital Readiness of Industrial Enterprises for Ecosystem Adaptatio...
SIG-955B81B681F1	2025-11-01	cordis.europa.eu	1	INTELLIGENT DIGITAL TOOLS FOR GREEN RENOVATION, ASSESSMENT, TRANSFORMATION, EFFICIENCY AND SUST...
SIG-3C2F7CAC7510	2025-10-01	openalex.org	1	Federated multi scale vision transformer with adaptive client aggregation for industrial defect...
SIG-6FBECA0F7EA7	2025-09-10	openalex.org	1	Design of a Sensor-Based Digital Product Passport for Low-Tech Manufacturing: Traceability and ...
SIG-1E3EF9A3D100	2025-09-01	openalex.org	1	An inceptive approach for designing simple digital twins and industrial metaverse process frame...
SIG-7697DB6F38B5	2025-08-30	openalex.org	1	The Green HACCP Approach: Advancing Food Safety and Sustainability
SIG-14947CE171FF	2026-08-14	stdaily.com	2	Traditional Turbine Factory Turns Intelligent
SIG-1E1C81947E78	2026-08-14	cbn.co.za	2	Electra Mining Africa showcases the technologies shaping tomorrow industry
SIG-5CEFDDFAA80A	2026-08-14	mining-technology.com	2	Why mining decarbonisation challenge starts with reliable connectivity
SIG-C1CD7564003C	2026-08-13	thisdaylive.com	2	AI , Digital Technology to Transform Energy Industry – THISDAYLIVE
SIG-1B97CEF269AB	2026-08-11	en.people.cn	2	N China Shanxi promotes green transformation of equipment manufacturing sector
SIG-288B4B508030	2026-08-10	arxiv.org	3	Automated generation of experimentally validated digital twins for desiccant-based low-dew-poin...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: competitive_landscape (no claim survived evidence binding); diagram (1 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:14+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:18+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:57:34+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.